

"Impact of Neuroplasticity-Enhancing Practices on Cognitive Engagement, Emotional Intelligence, and Academic Performance"

WORKFLOW OF DATA ANALYSIS (START → END)

1. Variable Selection for Analysis

Based on the research questions, variables were grouped into **three roles**: **A. Neuroplasticity-Enhancing Practices (Independent Variables)**

Used as **predictors** in regression analyses.

Variable name	Description
extracurricular_activities	Participation in extracurricular/professional activities
exercise_improve_cognition	Perceived cognitive benefits of physical exercise
sleep_hours	Average hours of uninterrupted sleep
nutritional_balanced_freq	Frequency of consuming balanced meals
nature_improve	Perceived cognitive benefit of nature exposure
creative_activities_benefits	Benefits of creative activities (art, music, writing)

B. Mediating Variables

Constructed as scale scores.

New Variable	Construct
cognitive_engagement	Cognitive engagement in learning
emotional_regulation	Emotional regulation / emotional intelligence

C. Outcome Variable

New Variable	Construct
academic_Performance	Learning habits and academic performance

Constructed as a scale score.

2. Recoding of Variables

Purpose of Recoding

- To convert categorical or textual responses into numeric form
- To ensure all Likert-type items follow the same direction
(higher value = higher level of the construct)

A. Likert-Scale Recoding Used

Label	Value
Strongly Disagree	1
Disagree	2
Neutral	3
Agree	4
Strongly Agree	5

Agreement scale

Label	1	2
Sometimes	3	
Rarely		
Often	4	
Always	5	

Frequency scale

Label	1	2	3	4
Not at all	5			
A little				
Neutral				
Mostly				

Exercise impact scale

Category	Value
Less than 4	1
4–5 hours	2
6–7 hours	3
8 or more	4

Sleep hours (ordinal numeric coding)

All recoding was done in SPSS → Transform → Recode into Different Variables, keeping original variables unchanged.

3. Scale Construction (Reliability & Composite Scores)

A. Cognitive Engagement Scale

Items used:

- cog_engage1
- cog_engage2
- cog_engage3
- cog_engage4

Procedure:

1. Analyze → Scale → Reliability Analysis
2. Cronbach's Alpha assessed
3. Reliability acceptable
4. Composite score created using Transform → Compute Variable
 - o Mean of retained items

5. New variable created: **cognitive_engagement**

B. Emotional Regulation Scale

Items used:

- exercise_regulate_emotion
- nutritional_mood_focus
- nature_improve

Procedure:

- Cronbach's Alpha checked
- Items retained based on reliability output
- Composite mean score computed
- New variable created: **emotional_regulation**

C. Academic Performance Scale

Items used:

- learning1 learning2
- performance_reflects_potential
- responsibilities_interfere
-

Procedure:

- Reliability analysis conducted
- Acceptable internal consistency
- Mean score computed
- New variable created: **academic_Performance**

4. Descriptive Statistics

Purpose

To summarize:

- Sample size (N)
- Mean level of each variable
- Variability (standard deviation)

Output Used

Analyze → Descriptive Statistics → Descriptives

How to Read the Table

- **Mean** :average level of the variable in the sample
- **Std. Deviation**: spread/variability of responses
- **Minimum–Maximum** :observed response range

This table provides an overview of students' engagement, emotional regulation, academic performance, and neuroplasticity practices.

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
extracurricular_activities	212	.00	5.00	.5142	.50098
exercise_improve_cognition	212	1.00		3.4009	1.08615
sleep_hours	212	1.00	4.00	2.6085	.89364
nutritional_balanced_freq	212	1.00	5.00	3.3396	1.06550
nature_improve	212	2	5	3.82	.933
creative_activities_benefits	212	1.00	5.00	3.9858	.83455
cognitive_engagement	212	2.00	5.00	3.7465	.65196
emotional_regulation	212	1.50	5.00	3.7901	.80109
academic_Performance	212	1.00	5.00	3.4858	.65340
Valid N (listwise)	212				

5. Correlation Analysis

Purpose

To examine bivariate relationships among:

- Neuroplasticity practices
- Mediators
- Academic performance

Procedure

Analyze → Correlate → Bivariate

Pearson correlation selected

Interpretation

- Positive correlation: variables increase together
- Significance ($p < .05$ / $p < .01$) indicates meaningful association

This step justified proceeding to regression and mediation analysis.

Correlations

		extracurricular_activities	exercise_improve_cognition
extracurricular_activities	Pearson Correlation	1	.273**
	Sig. (2-tailed)		.000
	N	212	212
exercise_improve_cognition	Pearson Correlation	.273**	1
	Sig. (2-tailed)	.000	
	N	212	212
nature_improve	Pearson Correlation	.244**	.232**
	Sig. (2-tailed)	.000	.001
	N	212	212
nutritional_balanced_freq	Pearson Correlation	.071	.148*
	Sig. (2-tailed)	.304	.031
	N	212	212
creative_activities_benefits	Pearson Correlation	.154*	.273**
	Sig. (2-tailed)	.025	.000
	N	212	212
sleep_hours	Pearson Correlation	-.025	-.072
	Sig. (2-tailed)	.722	.298
	N	212	212
cognitive_engagement	Pearson Correlation	.350**	.399**
	Sig. (2-tailed)	.000	.000
	N	212	212
emotional_regulation	Pearson Correlation	.264**	.378**
	Sig. (2-tailed)		
	N		

	Sig. (2-tailed)	.000	.000
	N	212	212
academic_Performance	Pearson Correlation	.319**	.183**
	Sig. (2-tailed)	.000	.007
	N	212	212

Correlations

		nature_improve	nutritional_balanced_freq	.071	.304
extracurricular_activities	Pearson Correlation	.244**	212	.148*	.031
	Sig. (2-tailed)	.000	212	.173*	.012
	N	212	212	1	
exercise_improve_cognition	Pearson Correlation	.232**			
	Sig. (2-tailed)	.001			
	N	212			
nature_improve	Pearson Correlation	1			
	Sig. (2-tailed)				
	N	212			
nutritional_balanced_freq	Pearson Correlation	.173*			
	Sig. (2-tailed)	.012			
	N	212	212		
creative_activities_benefits	Pearson Correlation	.270**		.112	
	Sig. (2-tailed)	.000		.104	
	N	212	212		
sleep_hours	Pearson Correlation	-.036		.011	
	Sig. (2-tailed)	.606		.875	
	N	212	212		
cognitive_engagement	Pearson Correlation	.355**		.143	
	Sig. (2-tailed)	.000		*	
	N	212		.037	
emotional_regulation	Pearson Correlation	.785**		212	
	Sig. (2-tailed)	.000		.151	
	N	212		*	
academic_Performance	Pearson Correlation	.217**		.028	
	Sig. (2-tailed)	.001		212	
	N	212		.129	
				.060	
Correlations				212	

		creative_activities _benefits	sleep_hours
extracurricular_activities	Pearson Correlation	.154*	-.025 .722
	Sig. (2-tailed)	.025	212 -.072
	N	212 .273** .000	.298 212
exercise_improve_cognition	Pearson Correlation	212	-.036 .606
	Sig. (2-tailed)	.270**	212 .011
	N	.000 212 .112	.875 212
nature_improve	Pearson Correlation	.104	-.084 .225
	Sig. (2-tailed)	212	212 1
	N	1	
nutritional_balanced_freq	Pearson Correlation		
	Sig. (2-tailed)		
	N		
creative_activities_benefits	Pearson Correlation		
	Sig. (2-tailed)		
	N	212	
sleep_hours	Pearson Correlation	-.084	
	Sig. (2-tailed)	.225	
	N	212	212
cognitive_engagement	Pearson Correlation	.313**	-.037
	Sig. (2-tailed)	.000	.593
	N	212	212
emotional_regulation	Pearson Correlation	.251**	.007
	Sig. (2-tailed)	.000	.918
	N	212	212
academic_Performance	Pearson Correlation	.210**	-.028
	Sig. (2-tailed)	.002	.687
	N	212	212

Correlations

		cognitive_engag ement	emotional_regula tion
extracurricular_activities	Pearson Correlation	.350**	.264**
	Sig. (2-tailed)	.000	.000
	N	212	212
exercise_improve_cognition	Pearson Correlation	.399**	.378**
	Sig. (2-tailed)	.000	.000

	N	212	212
nature_improve	Pearson Correlation	.355**	.785**
	Sig. (2-tailed)	.000	.000
	N	212	212
nutritional_balanced_freq	Pearson Correlation	.143*	.151
	Sig. (2-tailed)	.037	*
	N	212	.028
creative_activities_benefits	Pearson Correlation	.313**	212 **
	Sig. (2-tailed)	.000	.251
	N	212	.000
sleep_hours	Pearson Correlation	-.037	212
	Sig. (2-tailed)	.593	.007
	N	212	.918
cognitive_engagement	Pearson Correlation	1	212 ..
	Sig. (2-tailed)		.382
	N	212	.000
emotional_regulation	Pearson Correlation	.382**	212
	Sig. (2-tailed)	.000	1
	N	212	212
academic_Performance	Pearson Correlation	.405**	.243**
	Sig. (2-tailed)	.000	.000
	N	212	212

Correlations

		academic_Performance
extracurricular_activities	Pearson Correlation	.319**
	Sig. (2-tailed)	.000
	N	212
exercise_improve_cognition	Pearson Correlation	.183**
	Sig. (2-tailed)	.007
	N	212
nature_improve	Pearson Correlation	.217**
	Sig. (2-tailed)	.001
	N	212
nutritional_balanced_freq	Pearson Correlation	.129
	Sig. (2-tailed)	.060
	N	212

creative_activities_benefits	Pearson Correlation	.210**
	Sig. (2-tailed)	.002
	N	212
sleep_hours	Pearson Correlation	-.028
	Sig. (2-tailed)	.687
	N	212
cognitive_engagement	Pearson Correlation	.405**
	Sig. (2-tailed)	.000
	N	212
emotional_regulation	Pearson Correlation	.243**
	Sig. (2-tailed)	.000
	N	212
academic_Performance	Pearson Correlation	1
	Sig. (2-tailed)	
	N	212

** . Correlation is significant at the 0.01 level (2-tailed).

* . Correlation is significant at the 0.05 level (2-tailed).

6. Regression Analyses (Research Question–Wise)

Research Question 1

How are neuroplasticity-enhancing practices associated with cognitive engagement?

Analysis

- **DV:** cognitive_engagement
- **IVs:** all neuroplasticitypractices
- Method: Multiple Linear Regression (Enter)

Interpretation

The regression model was statistically meaningful, explaining approximately **29.6%** of the variance in cognitive engagement. Extracurricular activities, exercise that improves cognition, nature exposure, and creative activities emerged as significant positive predictors of cognitive engagement. Sleep hours and balanced nutrition did not show significant direct effects in this model.

Conclusion

Neuroplasticity-enhancing practices significantly predicted cognitive engagement, with exercise, extracurricular activities, nature exposure, and creative activities emerging as meaningful contributors.

Model Summaryb				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.544a	.296	.276	.55480

a. Predictors: (Constant), creative_activities_benefits, sleep_hours, nutritional_balanced_freq, extracurricular_activities, nature_improve, exercise_improve_cognition

b. Dependent Variable: cognitive_engagement

ANOVAa						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	26.584	6	4.431	14.395	.000b
	Residual	63.100	205	.308		
	Total	89.685	211			

a. Dependent Variable: cognitive_engagement

b. Predictors: (Constant), creative_activities_benefits, sleep_hours, nutritional_balanced_freq, extracurricular_activities, nature_improve, exercise_improve_cognition

Coefficientsa						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.994	.271 .081		7.346	.000
	extracurricular_activities	.270	.037 .044	.208	3.343	.001
	nutritional_balanced_freq	.025	.038	.040	.669	.504
	nature_improve	.138		.198	3.130	.002
	exercise_improve_cognition	.149		.248	3.892	.000
	sleep_hours	.004	.043	.006	.096	.923
	creative_activities_benefits	.122	.049	.156	2.494	.013

Coefficientsa			
Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	extracurricular_activities	.889	1.124
	nutritional_balanced_freq	.955	1.047
	nature_improve	.857	1.167
	exercise_improve_cognition	.847	1.181
	sleep_hours	.990	1.011
	creative_activities_benefits	.873	1.146

a. Dependent Variable: cognitive_engagement

Collinearity Diagnosticsa						
Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	extracurricular_activities	nutritional_balanced_freq
1	1	6.280	1.000	.00	.01	.00

	2	.425	3.844	.00	.90	.01
	3	.110	7.551	.00	.04	.04
	4	.078	9.001	.00	.02	.78
	5	.057	10.505	.01	.00	.12
	6	.035	13.307	.02	.02	.00
	7	.015	20.175	.97	.01	.05

Collinearity Diagnostics ^a							
Model	Dimension	Variance Proportions					
		nature_improve	exercise_improve _cognition	sleep_hours	creative_activities _benefits	.00	.00
1	1	.00	.00	.00	.01	.01	.08 .50
	2	.00	.00 .11 .27 .55	.01	.40		
	3	.01	.05 .02	.69			
	4	.00		.00			
	5	.28		.07			
	6	.61		.01			
	7	.09		.21			

a. Dependent Variable: cognitive_engagement

Residuals Statistics ^a					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	2.9147	4.4416	3.7465	.35495	212
Residual	-1.36437	1.51834	.00000	.54686	212
Std. Predicted Value	-2.343	1.958	.000	1.000	212
Std. Residual	-2.459	2.737	.000	.986	212

a. Dependent Variable: cognitive_engagement

Research Question 2

What is the relationship between neuroplasticity-enhancing practices and emotional intelligence?

Analysis

- DV: emotional_regulation
- IVs: all neuroplasticity practices

Interpretation The model explained a substantial proportion of variance in emotional regulation ($R^2 = .660$). Among the predictors, nature exposure demonstrated a very strong and significant positive effect, followed by exercise improving cognition. Other practices did not contribute significantly when all variables were considered simultaneously. **Conclusion** Nature exposure and exercise showed strong positive associations with emotional regulation, indicating their importance in emotional self-regulation processes.

Model Summary ^b					
Model	R	R Square	Adjusted R Square	R	Std. Error of the Estimate
1	.812a	.660	.650		.47416

a. Predictors: (Constant), creative_activities_benefits, sleep_hours, nutritional_balanced_freq, extracurricular_activities, nature_improve, exercise_improve_cognition

b. Dependent Variable: emotional_regulation

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	89.320	6	14.887	66.214	.000b
	Residual	46.089	205	.225		
	Total	135.409	211			

a. Dependent Variable: emotional_regulation

b. Predictors: (Constant), creative_activities_benefits, sleep_hours, nutritional_balanced_freq, extracurricular_activities, nature_improve, exercise_improve_cognition

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	.769	.232	.069	3.317	.001
	extracurricular_activities	.051	.031	.038	.742	.459
	nutritional_balanced_freq	-.007	.033	-.009	-.212	.832
	nature_improve	.630		.734	16.664	.000
	exercise_improve_cogniti	.151		.204	4.617	.000

	on					
	sleep_hours	.044	.037	.049	1.186	.237
	creative_activities_benefits	-.003	.042	-.003	-.076	.939

Coefficients ^a			
Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	extracurricular_activities	.889	1.124
	nutritional_balanced_freq	.955	1.047
	nature_improve	.857	1.167
	exercise_improve_cognition	.847	1.181
	sleep_hours	.990	1.011
	creative_activities_benefits	.873	1.146

a. Dependent Variable: emotional_regulation

Collinearity Diagnostics ^a						
Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	extracurricular_activities .01	nutritional_balanced_freq
1	1	6.280	1.000	.00	.90 .04 .02 .00	.00 .01 .04
	2	.425	3.844	.00	.02 .01	.78 .12 .00
	3	.110	7.551	.00		.05
	4	.078	9.001	.00		
	5	.057	10.505	.01		
	6	.035	13.307	.02		
	7	.015	20.175	.97		

Collinearity Diagnostics ^a					
Model	Dimension	Variance Proportions			
		nature_improve	exercise_improve_cognition .00 .00	sleep_hours	creative_activities_benefits .00 .00
1	1	.00	.11 .27	.00	.01 .01
	2	.00		.01	
	3	.01		.69	
	4	.00		.00	

	5 .28 6 .61 7 .09		.55	.07	.08
			.05	.01	.50
			.02	.21	.40

a. Dependent Variable: emotional_regulation

Residuals Statisticsa					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	2.3109	4.8644	3.7901	.65063	212
Residual	-1.26083	.97575	.00000	.46737	212
Std. Predicted Value	-2.274	1.651	.000	1.000	212
Std. Residual	-2.659	2.058	.000	.986	212

a. Dependent Variable: emotional_regulation

Research Question 3

To what extent do cognitive engagement and emotional intelligence predict academic performance?

Analysis

- **DV:** academic_Performance
- **IVs:** cognitive_engagement, emotional_regulation

Interpretation

The regression model explained of the 17.31% variance in academic performance. Cognitive engagement was a significant positive predictor, whereas emotional regulation did not reach statistical significance when both variables were included together.

Conclusion

Cognitive engagement significantly predicted academic performance, whereas emotional regulation did not show a significant direct effect when both were included.

Model Summaryb				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.416a	.173	.165	.59707

a. Predictors: (Constant), cognitive_engagement, emotional_regulation

b. Dependent Variable: academic_Performance

ANOVAa						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	15.576	2	7.788	21.846	.000b
	Residual	74.507	209	.356		
	Total	90.083	211			

a. Dependent Variable: academic_Performance

b. Predictors: (Constant), cognitive_engagement, emotional_regulation

Coefficientsa						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.795	.265 .056		6.773	.000
	emotional_regulation	.085	.068	.104	1.527	.128
	cognitive_engagement	.366		.365	5.360	.000

Coefficientsa		
Model		Collinearity Statistics
		Tolerance
1	(Constant)	
	emotional_regulation	.854
	cognitive_engagement	.854

a. Dependent Variable: academic_Performance

Collinearity Diagnosticsa						
Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	emotional_regulation	cognitive_engagement .00
1	1	2.961	1.000	.00	.00	.20 .79
	2	.024	11.009	.13	.99 .01	
	3	.015	14.189	.86		

a. Dependent Variable: academic_Performance

Residuals Statisticsa					
	Minimum	Maximum	Mean	Std. Deviation	N

Predicted Value	2.7378	4.0468	3.4858	.27169	212
Residual	-2.46921	1.76216	.00000	.59423	212
Std. Predicted Value	-2.753	2.065	.000	1.000	212
Std. Residual	-4.136	2.951	.000	.995	212

a. Dependent Variable: academic_Performance

Research Question 4

Do cognitive engagement and emotional intelligence mediate the effect of neuroplasticity practices on academic performance?

Mediation Steps Followed

1. Neuroplasticity practices → Academic performance (direct effect)
2. Neuroplasticity practices → Mediators
3. Mediators → Academic performance

Interpretation

The regression model indicated that neuroplasticity-enhancing practices explained **14.8%** of the variance in academic performance. Extracurricular activities emerged as a significant direct predictor, while other practices showed weaker or non-significant direct effects. When cognitive engagement was introduced in subsequent models, the direct effects of several practices were reduced.

Conclusion

Cognitive engagement partially mediated the relationship between neuroplasticity practices and academic performance. Emotional regulation did not function as a mediator.

Model Summary ^b				
Model	R	R Square	Adjusted R Square	R
1	.384 ^a	.148	.123	

a. Predictors: (Constant), extracurricular_activities, sleep_hours, nutritional_balanced_freq, creative_activities_benefits, nature_improve, exercise_improve_cognition

b. Dependent Variable: academic_Performance

ANOVAa						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	13.302	6	2.217	5.919	.000b
	Residual	76.780	205	.375		
	Total	90.083	211			

a. Dependent Variable: academic_Performance

b. Predictors: (Constant), extracurricular_activities, sleep_hours, nutritional_balanced_freq, creative_activities_benefits, nature_improve, exercise_improve_cognition

Coefficientsa						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.434	.299	.047	8.129	.000
	sleep_hours	-.004	.054	-.005	-.080	.936
	creative_activities_benefits	.097		.123	1.786	.076
	nutritional_balanced_freq	.045	.040	.074	1.120	.264
	nature_improve	.068	.049	.097	1.397	.164
	exercise_improve_cognition	.027	.042	.045	.646	.519
	extracurricular_activities	.338	.089	.259	3.785	.000

Coefficientsa			
Model		Collinearity Statistics	
		Tolerance	VIF
1	(Constant)		
	sleep_hours	.990	1.011
	creative_activities_benefits	.873	1.146
	nutritional_balanced_freq	.955	1.047
	nature_improve	.857	1.167
	exercise_improve_cognition	.847	1.181
	extracurricular_activities	.889	1.124

a. Dependent Variable: academic_Performance

Collinearity Diagnosticsa

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	sleep_hours	creative_activities_benefits
1	1	6.280	1.000	.00	.00	.00 .00 .01
	2	.425	3.844	.00	.01	.01
	3	.110	7.551	.00	.69	.08
	4	.078	9.001	.00	.00	.50
	5	.057	10.505	.01	.07	.40
	6	.035	13.307	.02	.01	
	7	.015	20.175	.97	.21	

Collinearity Diagnosticsa															
Model	Dimension	Variance Proportions													
		nutritional_balanced_freq			nature_improve		exercise_improve_cognition			extracurricular_activities					
1	1	.78	.12	.00	.05	.00		.00	.11	.27	.55	.04	.02	.00	.02
	2					.00		.05	.02			.01			
	3					.01									
	4					.00									
	5					.28									
	6					.61									
	7					.09									

a. Dependent Variable: academic_Performance

Residuals Statisticsa					
	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	2.9135	3.9532	3.4858	.25109	212
Residual	-2.25373	1.64213	.00000	.60323	212
Std. Predicted Value	-2.280	1.861	.000	1.000	212
Std. Residual	-3.683	2.683	.000	.986	212

a. Dependent Variable: academic_Performance

Research Question 5

Which neuroplasticity-related behaviors are the strongest predictors of academic performance?

Interpretation

Among neuroplasticity-enhancing practices, extracurricular activities were the strongest direct predictor of academic performance. Additionally, exercise, nature exposure, and creative activities indirectly influenced academic performance through increased cognitive engagement.

Conclusion

Extracurricular activities emerged as the strongest direct predictor of academic performance, while exercise, nature exposure, and creative activities influenced performance indirectly through cognitive engagement.

Final Summary

The analysis demonstrates that **neuroplasticity-enhancing behaviors play a meaningful role in students' academic outcomes**, primarily through **cognitive engagement**, highlighting the importance of active, enriching, and health-promoting practices in university life.